

Lesson 23: Introduction to Bayesian Inference

BSTA 551: Statistical Inference

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2026-03-11

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Today's Goals

1. Understand the **Bayesian paradigm**: parameters as random variables with prior distributions
2. Derive **posterior distributions** using Bayes' theorem for random variables
3. Work with **conjugate prior** families: Beta/Binomial, Gamma/Exponential, Gamma/Poisson, Normal/Normal
4. Compute **Bayes estimators** (posterior mean, MAP) and understand their properties
5. Construct **Bayesian credible intervals** and compare with frequentist confidence intervals
6. Perform **Bayesian hypothesis testing** via posterior probabilities

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Reading:

- Devore, Berk, and Carlton, Chapter 15 (Introduction to Bayesian Estimation)
- Chihara and Hesterberg, Chapter 10 (Bayesian Methods)

Frequentist vs. Bayesian Inference

Two Paradigms for Statistical Inference

Frequentist (Classical) Inference:

Observe data $X \sim f(x|\theta) \longrightarrow$ use observed data x to make statements about θ , which is assumed to be a **fixed (and unknown)** model parameter.

- Point estimates, confidence intervals, hypothesis tests
- θ is a constant; probability statements are about **the data**, not θ
- Confidence interval interpretation: long-run frequency of coverage

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Bayesian Inference:

Model random variable $\theta \sim \pi(\theta) \longrightarrow$ observe $X|\theta \sim f(x|\theta) \longrightarrow$ update with $\pi(\theta|x)$

- θ is treated as a **random variable** with a probability distribution
- **Prior distribution** $\pi(\theta)$: incorporates available information about θ before data are observed
- **Posterior distribution** $\pi(\theta|x)$: “updated” distribution of θ after the data are observed
- All inference comes from the posterior

Why Consider Bayesian Methods?

Advantages:

- Incorporates prior knowledge or expert opinion formally
- Posterior probability statements are direct: $P(\theta \in A|x)$ has a natural interpretation
- Credible intervals say “there is a 95% probability that θ is in this interval”
- Naturally handles nuisance parameters through marginalization
- Provides a coherent framework for sequential updating of evidence

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Challenges:

- Requires specifying a prior distribution (can be subjective)
- Choice of prior can strongly influence results, especially with small samples
- Computational demands can be substantial for complex models
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Medical research context: Bayesian methods are increasingly used in clinical trials (adaptive designs), diagnostic testing, meta-analysis, and regulatory submissions (e.g., medical device trials at the FDA).

Prior and Posterior Distributions

Prior Distributions

Definition: Prior Distribution (DBC 15.1)

A **prior distribution** for a parameter θ , denoted $\pi(\theta)$, is a probability distribution on the set of possible values for θ . If θ takes values in an interval I , then $\pi(\theta)$ is a pdf satisfying $\int_I \pi(\theta) d\theta = 1$. If θ takes values in a discrete set D , then $\pi(\theta)$ is a pmf satisfying $\sum_{\theta \in D} \pi(\theta) = 1$.

Prior Distributions

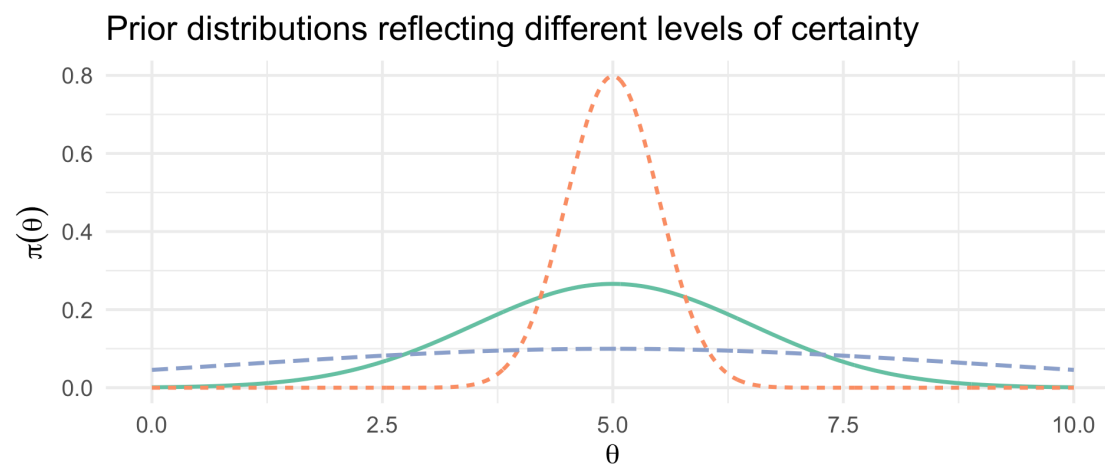
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The **variability** of the prior distribution reflects how certain we are about our initial belief:

- A **narrow** (concentrated) prior expresses strong belief
- A **wide** (diffuse) prior expresses substantial uncertainty
- A **flat** or **noninformative** prior attempts to let the data speak for themselves

► Code



The Posterior Distribution

The **key** to Bayesian inference is Bayes' theorem applied to random variables.

Definition: Posterior Distribution

Suppose $X_1, \dots, X_n$ have joint pdf $f(x_1, \dots, x_n; \theta)$ and θ has been assigned a prior distribution $\pi(\theta)$. The **posterior distribution** of θ given the observed data is:

$$\pi(\theta|x_1, \dots, x_n) = \frac{\pi(\theta) f(x_1, \dots, x_n; \theta)}{\int \pi(\theta) f(x_1, \dots, x_n; \theta) d\theta}$$

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The denominator $m(x) = \int \pi(\theta) f(x|\theta) d\theta$ is the **marginal distribution** of the data. Since it does not depend on θ , we often write:

$$\pi(\theta|x) \propto \underbrace{\pi(\theta)}_{\text{prior}} \times \underbrace{f(x|\theta)}_{\text{likelihood}}$$

$$\boxed{\text{Posterior} \propto \text{Prior} \times \text{Likelihood}}$$

This is the fundamental equation of Bayesian inference.

Bayesian Updating: The Full Picture

The complete set of distributions in a Bayesian analysis:

1. **Prior distribution:** $\theta \sim \pi(\theta)$ (set in advance)
2. **Data model (likelihood):** $X|\theta \sim f(x|\theta)$ (set in advance)
3. **Joint distribution:** $f_{X,\theta}(x, \theta) = f(x|\theta) \pi(\theta)$
4. **Marginal distribution:** $m(x) = \int f(x|\theta) \pi(\theta) d\theta$
5. **Posterior distribution:** $\pi(\theta|x) = \frac{f(x|\theta) \pi(\theta)}{m(x)}$

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The posterior combines **prior belief** with **observed evidence** from the data. As we collect more data, the posterior becomes increasingly driven by the likelihood.

Conjugate Priors

What Are Conjugate Priors?

In practice, we want to choose a prior distribution family such that the posterior distribution belongs to the **same family**. When this happens, we say the prior is **conjugate** to the data distribution.

Definition: Conjugate Prior

A prior distribution $\pi(\theta)$ from family $\mathcal{F}$ is **conjugate** to the data distribution $f(x|\theta)$ if the posterior distribution $\pi(\theta|x)$ also belongs to $\mathcal{F}$.

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- Posterior has a **closed-form** expression — no numerical integration needed
- The prior hyperparameters have interpretable roles (e.g., “prior sample size”)
- Easy to update sequentially as new data arrive

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Common conjugate pairs:

Data Model	Conjugate Prior	Posterior
Bernoulli/Binomial(p)	Beta(α_0, β_0)	Beta($\alpha_0 + y, \beta_0 + n - y$)
Poisson(λ)	Gamma(α_0, β_0)	Gamma($\alpha_0 + \sum x_i, \frac{1}{n+1/\beta_0}$)
Exponential(λ)	Gamma(α_0, β_0)	Gamma($\alpha_0 + n, \frac{1}{1/\beta_0 + \sum x_i}$)

Example: Beta Prior for Binomial Data (DBC 15.3, C&H 10.3)

Proposition: Beta-Binomial Conjugacy

Let $X_1, \dots, X_n$ be a random sample from a Bernoulli(p) distribution. Equivalently, let $Y = \sum X_i \sim \text{Bin}(n, p)$. If p is assigned a $\text{Beta}(\alpha_0, \beta_0)$ prior, then the posterior distribution of p given the data is:

$$p \mid y \sim \text{Beta}(\alpha_0 + y, \beta_0 + n - y)$$

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$$p \mid y \sim \text{Beta}(\alpha_0 + y, \beta_0 + n - y)$$

Proof: The joint Bernoulli pmf is $f(x|p) = p^y(1 - p)^{n-y}$ where $y = \sum x_i$. The Beta prior is $\pi(p) \propto p^{\alpha_0-1}(1 - p)^{\beta_0-1}$. Applying Bayes' theorem:

$$\pi(p|x) \propto p^{\alpha_0-1}(1 - p)^{\beta_0-1} \cdot p^y(1 - p)^{n-y} = p^{(\alpha_0+y)-1}(1 - p)^{(\beta_0+n-y)-1}$$

We recognize this as proportional to the Beta($\alpha_0 + y, \beta_0 + n - y$) density. $\square$

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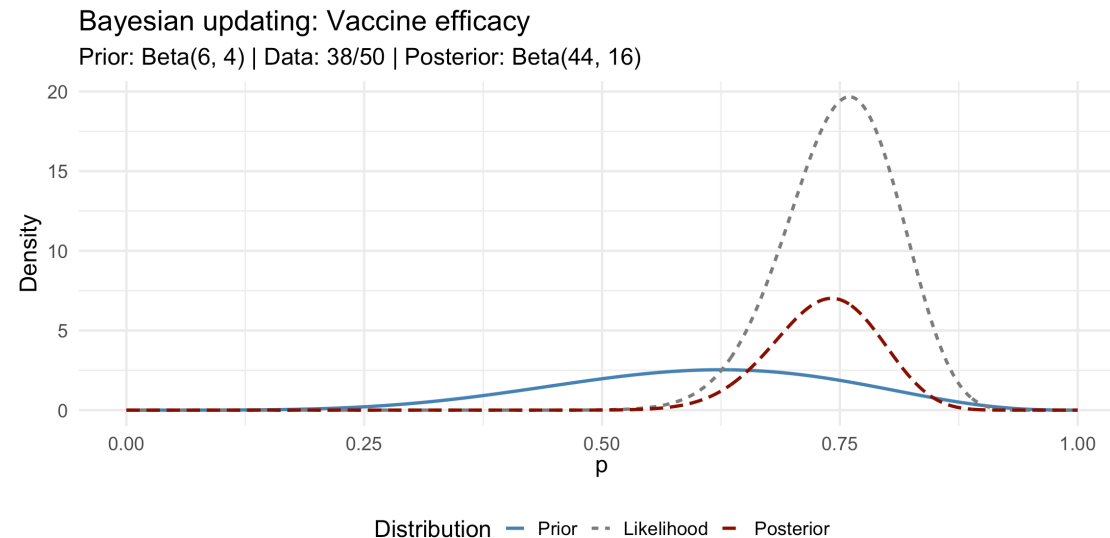
Interpretation: The hyperparameters α_0 and β_0 act like “prior successes” and “prior failures.” The posterior adds observed successes y and observed failures $n - y$.

Medical Example: Beta-Binomial — Vaccine Efficacy

A Phase II clinical trial tests a new vaccine. Prior studies suggest efficacy around 60%, but with substantial uncertainty. We use a $\text{Beta}(6, 4)$ prior for p (the true efficacy rate), which has mean $6/10 = 0.60$.

In the trial, $n = 50$ patients are vaccinated and $y = 38$ show a positive immune response.

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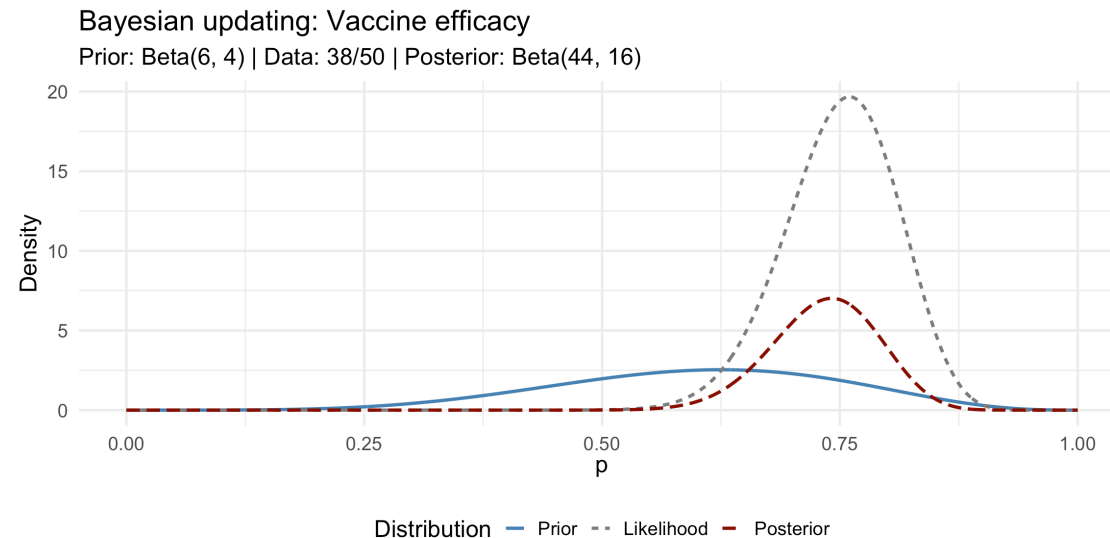


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- Prior mean: $\frac{6}{10} = 0.60$
- MLE (frequentist): $\hat{p} = 38/50 = 0.76$
- Posterior mean: $\frac{44}{60} = 0.7333$ – a weighted average, pulled toward the data

Example: Gamma Prior for Exponential Data (DBC 15.2)

Proposition: Gamma-Exponential Conjugacy

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exponential}(\lambda)$ with pdf $f(x|\lambda) = \lambda e^{-\lambda x}$. If λ is assigned a $\text{Gamma}(\alpha_0, \beta_0)$ prior, the posterior distribution of λ is:

$$\lambda \mid x_1, \dots, x_n \sim \text{Gamma}\left(\alpha_0 + n, \frac{1}{1/\beta_0 + \sum x_i}\right)$$

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Proof: The joint likelihood is $f(x|\lambda) = \lambda^n e^{-\lambda \sum x_i}$. The Gamma prior is $\pi(\lambda) \propto \lambda^{\alpha_0-1} e^{-\lambda/\beta_0}$. Then:

$$\pi(\lambda|x) \propto \lambda^{\alpha_0-1} e^{-\lambda/\beta_0} \cdot \lambda^n e^{-\lambda \sum x_i} = \lambda^{(\alpha_0+n)-1} \exp\left(-\lambda \left(\frac{1}{\beta_0} + \sum x_i\right)\right)$$

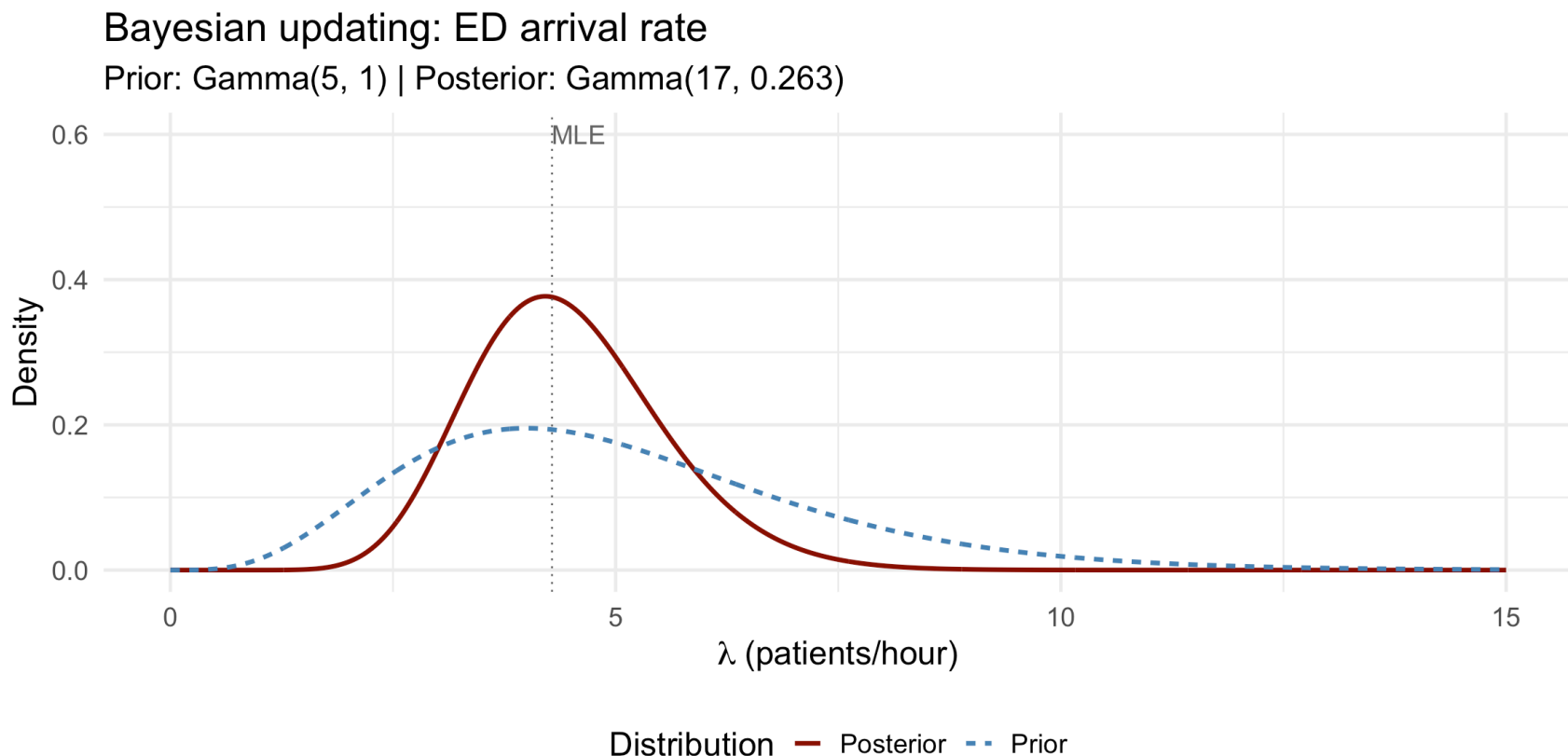
This is the kernel of a Gamma distribution with shape $\alpha_0 + n$ and scale $\frac{1}{1/\beta_0 + \sum x_i}$. $\square$

Medical Example: Gamma-Exponential – Hospital Wait Times

An emergency department models the rate λ (patients per hour) of arrivals. Prior experience suggests approximately 5 patients per hour, so we use a $\text{Gamma}(5, 1)$ prior for λ (mean = 5).

We observe $n = 12$ inter-arrival times (in hours) with $\sum x_i = 2.8$.

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Example: Gamma Prior for Poisson Data (C&B, C&H 10.4)

Proposition: Gamma-Poisson Conjugacy

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\theta)$. If θ is assigned a $\text{Gamma}(\alpha_0, \beta_0)$ prior, then:

$$\theta \mid x_1, \dots, x_n \sim \text{Gamma}\left(\alpha_0 + \sum_{i=1}^n x_i, \frac{1}{n + 1/\beta_0}\right)$$

Proof: The joint pmf is $f(x|\theta) = \frac{\theta^{\sum x_i} e^{-n\theta}}{\prod x_i!}$. Combining with the Gamma prior $\pi(\theta) \propto \theta^{\alpha_0-1} e^{-\theta/\beta_0}$:

$$\pi(\theta|x) \propto \theta^{\alpha_0-1} e^{-\theta/\beta_0} \cdot \theta^{\sum x_i} e^{-n\theta} = \theta^{(\alpha_0 + \sum x_i)-1} \exp\left(-\theta \left(n + \frac{1}{\beta_0}\right)\right)$$

which is the $\text{Gamma}\left(\alpha_0 + \sum x_i, \frac{1}{n + 1/\beta_0}\right)$ kernel. $\square$

Example: Normal Prior for Normal Mean (DBC 15.9, C&H 10.4)

Proposition: Normal-Normal Conjugacy

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known. If μ is assigned a $N(\mu_0, \sigma_0^2)$ prior, then:

$$\mu \mid x_1, \dots, x_n \sim N(\mu_1, \sigma_1^2)$$

where the **posterior precision** (reciprocal of variance) is:

$$\frac{1}{\sigma_1^2} = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}$$

and the **posterior mean** is a weighted average:

$$\mu_1 = \frac{\frac{1}{\sigma_0^2}}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}} \cdot \mu_0 + \frac{\frac{n}{\sigma^2}}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}} \cdot \bar{x}$$

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Key insight — Precision adds:

$$\underbrace{\frac{1}{\sigma_1^2}}_{\text{posterior precision}} = \underbrace{\frac{1}{\sigma_0^2}}_{\text{prior precision}} + \underbrace{\frac{n}{\sigma^2}}_{\text{data precision}}$$

The weights in the posterior mean are proportional to the respective precisions!

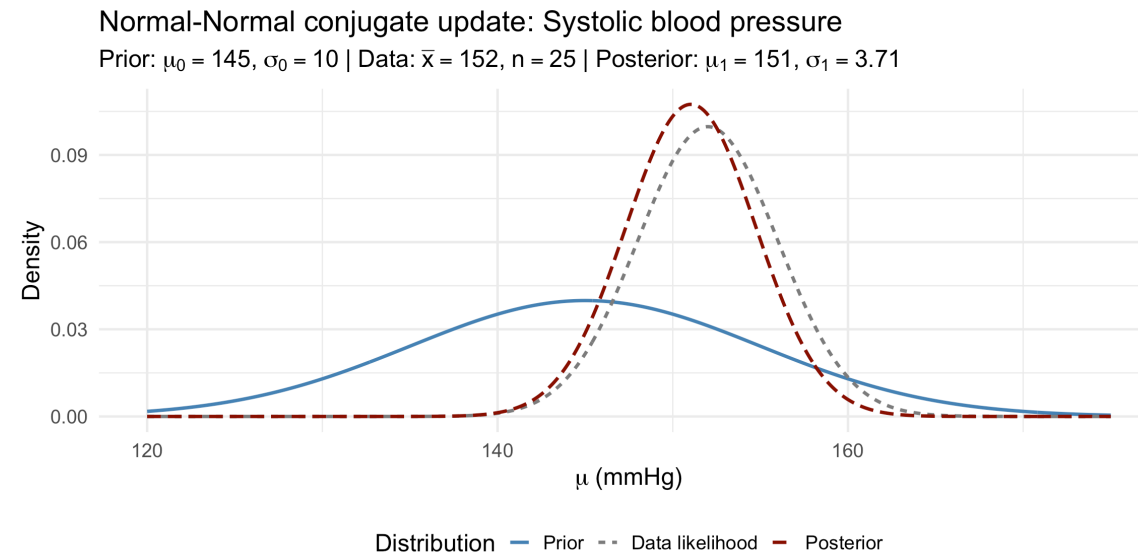


Medical Example: Normal-Normal – Systolic Blood Pressure

A researcher studies the mean systolic blood pressure μ in a hypertensive population. Prior clinical knowledge suggests $\mu \approx 145$ mmHg with $\sigma_0 = 10$. Population $\sigma = 20$ mmHg is assumed known.

A sample of $n = 25$ patients has $\bar{x} = 152$ mmHg.

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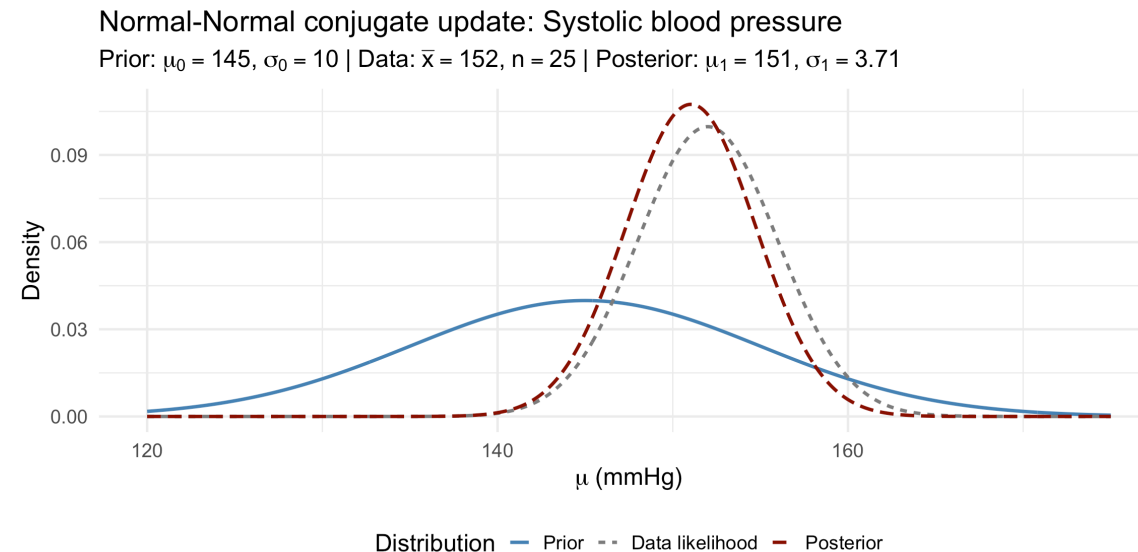


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- **Prior mean:** $\mu_0 = 145$, **Data mean:** $\bar{x} = 152$
- **Posterior mean:** $\mu_1 = 151.03$ – between the two, weighted by precision
- **Posterior SD:** $\sigma_1 = 3.71$ – always **smaller** than both σ_0 and $\sigma/\sqrt{n}$

Proof: Normal-Normal Conjugacy

Proof sketch: Apply the posterior formula with $\pi(\mu) = \frac{1}{\sqrt{2\pi}\sigma_0} e^{-(\mu-\mu_0)^2/(2\sigma_0^2)}$ and the normal likelihood:

$$\pi(\mu|x) \propto \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right) \prod_{i=1}^n \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Expanding the exponent and collecting terms in μ :

$$\pi(\mu|x) \propto \exp\left(-\frac{1}{2} \left[\frac{(\mu - \mu_0)^2}{\sigma_0^2} + \frac{\sum (x_i - \mu)^2}{\sigma^2} \right]\right)$$

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Completing the square in μ (the key algebraic step):

$$= \exp\left(-\frac{(\mu - \mu_1)^2}{2\sigma_1^2} + C\right)$$

where C does not depend on μ , and μ_1, σ_1^2 are as defined in the proposition.

Since this is proportional to a $N(\mu_1, \sigma_1^2)$ density in μ , the posterior must be $N(\mu_1, \sigma_1^2)$. $\square$

Bayesian Point Estimation

The Bayes Estimator

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The most common Bayesian point estimator of θ is the **mean of the posterior distribution**:

$$\hat{\theta}_B = E(\theta \mid X_1, \dots, X_n)$$

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For our conjugate families:

Model	Bayes Estimator (Posterior Mean)
Beta($\alpha_0 + y, \beta_0 + n - y$)	$\hat{p}_B = \frac{\alpha_0 + y}{\alpha_0 + \beta_0 + n}$
Gamma($\alpha_0 + n, \frac{1}{1/\beta_0 + \sum x_i}$)	$\hat{\lambda}_B = \frac{\alpha_0 + n}{1/\beta_0 + \sum x_i}$
$N(\mu_1, \sigma_1^2)$	$\hat{\mu}_B = \mu_1$ (weighted average of μ_0 and $\bar{x}$)

The Bayes Estimator as a Weighted Average

For the **Binomial-Beta** case, the Bayes estimator can be written as:

$$\hat{p}_B = \frac{\alpha_0 + y}{\alpha_0 + \beta_0 + n} = \underbrace{\frac{\alpha_0 + \beta_0}{\alpha_0 + \beta_0 + n}}_{\text{weight on prior}} \cdot \underbrace{\frac{\alpha_0}{\alpha_0 + \beta_0}}_{\text{prior mean}} + \underbrace{\frac{n}{\alpha_0 + \beta_0 + n}}_{\text{weight on data}} \cdot \underbrace{\frac{y}{n}}_{\text{sample proportion}}$$

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As $n \rightarrow \infty$: the weight on the prior $\rightarrow 0$ and the weight on the data $\rightarrow 1$.

$$\hat{p}_B \rightarrow \frac{y}{n} = \hat{p}_{MLE}$$

The data dominates the prior for large samples.

Properties of Bayes Estimators

Under general conditions, as $n \rightarrow \infty$:

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These properties imply that **Bayes estimators are generally consistent** (in the frequentist sense).

Key consequence: For large samples, Bayesian and frequentist estimates will typically be quite close, regardless of the prior chosen (as long as the prior is not degenerate — i.e., assigns zero probability to the true value).

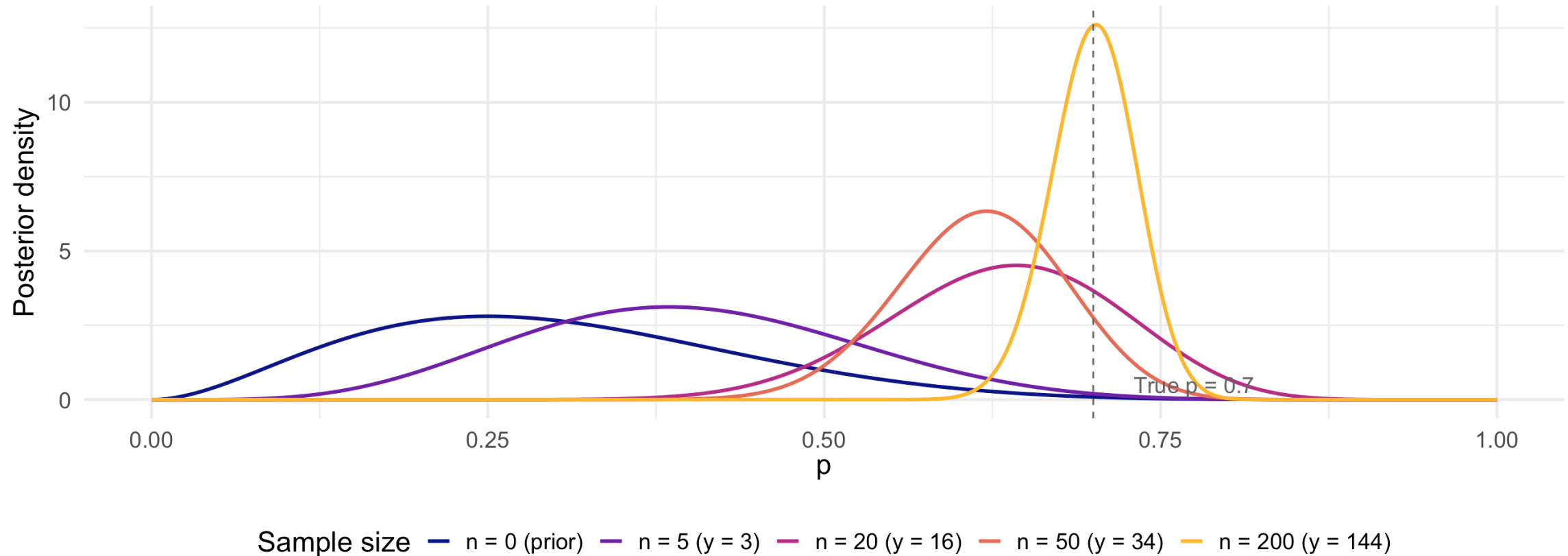
For small samples, the prior can have a substantial influence. This is a feature (incorporating prior knowledge) or a bug (sensitivity to subjective choices), depending on one's perspective.

Simulation: Effect of Sample Size on Prior Influence

► Code

Posterior distribution evolves toward the truth as n increases

Prior: Beta(3, 7), centered at 0.3 — far from the true $p = 0.7$



Bayesian Hypothesis Testing

Bayesian Approach to Hypothesis Testing

In frequentist testing, θ is fixed and a hypothesis is either true or false. The probabilities $P(H_0 \text{ is true}|x)$ and $P(H_1 \text{ is true}|x)$ are simply 0 or 1.

In Bayesian testing, θ has a distribution, so we can **directly compute**:

$$P(H_0 \text{ is true}|x) = P(\theta \in \Theta_0|x) = \int_{\Theta_0} \pi(\theta|x) d\theta$$

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Decision rules (examples):

- Accept H_0 if $P(\theta \in \Theta_0|x) \geq P(\theta \in \Theta_0^c|x)$
- Accept H_0 if $P(\theta \in \Theta_0|x) > 0.95$ (or some high threshold)

Medical Example: Bayesian Test – Drug Response Rate

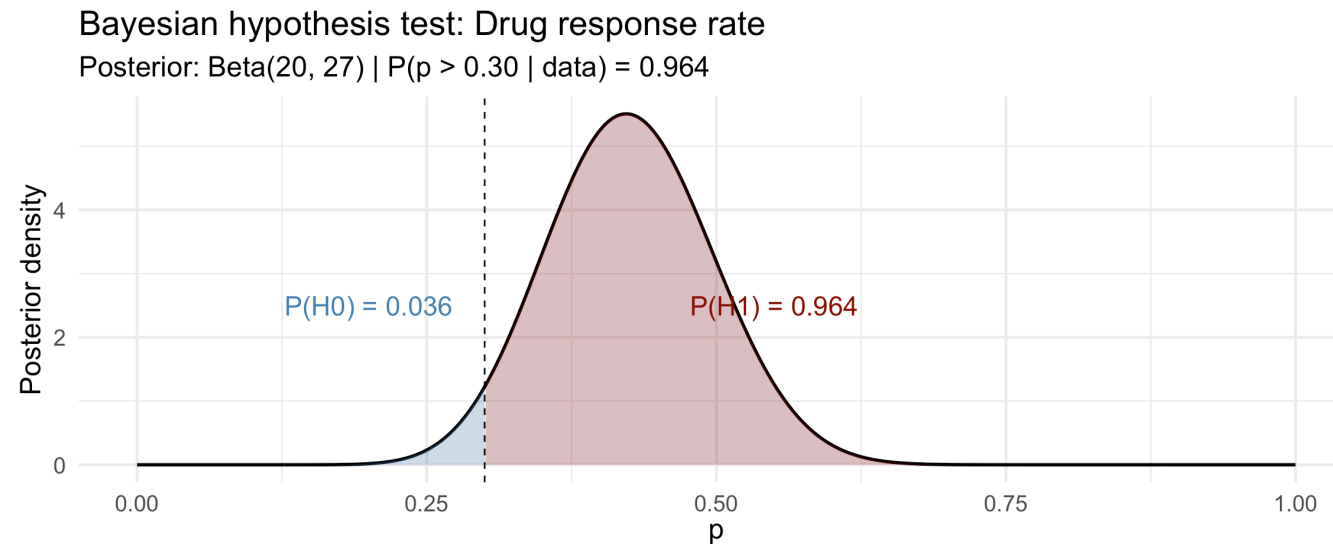
A new cancer drug is tested. We want to assess:

$$H_0 : p \leq 0.30 \quad \text{vs} \quad H_1 : p > 0.30$$

where p is the response rate. We use a Beta(2, 5) prior (prior mean ≈ 0.29 , reflecting skepticism).

In the trial, $n = 40$ patients are treated and $y = 18$ respond.

► Code



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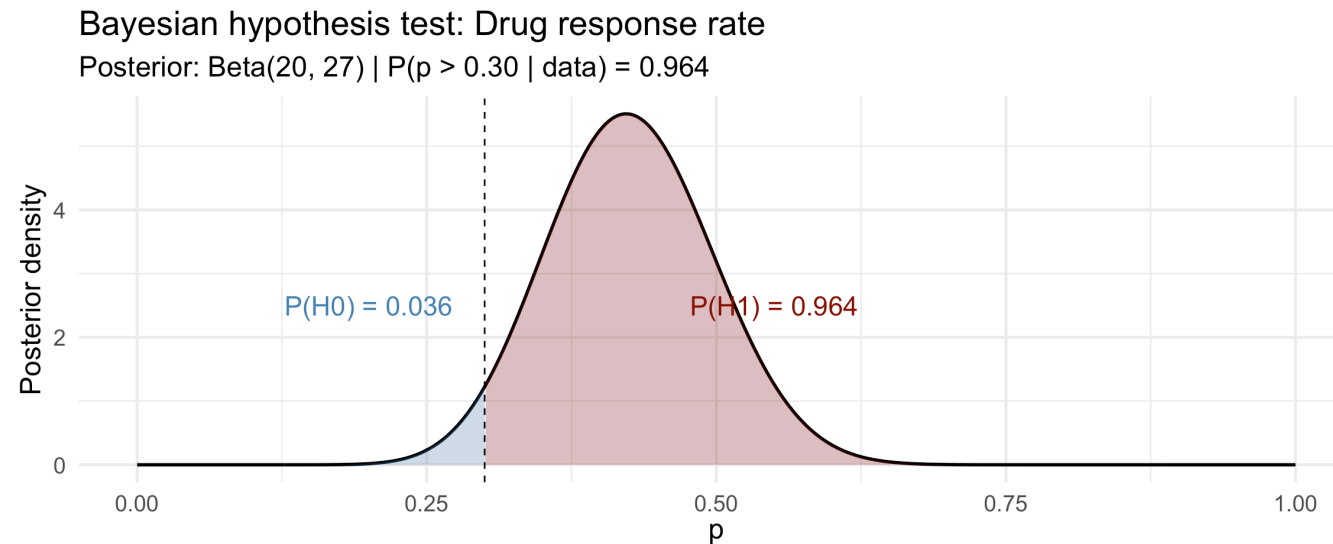
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In the trial, $n = 40$ patients are treated and $y = 18$ respond.

► Code



The posterior probability that $p > 0.30$ is 0.964, providing strong evidence that the drug's response rate exceeds 30%.

Example: Bayesian Test with Bernoulli Data (C&B 8.2.2)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$ with prior $p \sim \text{Uniform}(0, 1) = \text{Beta}(1, 1)$.

$$H_0 : p \leq 1/2 \quad \text{vs} \quad H_1 : p > 1/2$$

The posterior is $\text{Beta}(\sum x_i + 1, n - \sum x_i + 1)$.

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Case 1: $n = 20, \sum x_i = 11$. Then $p \mid x \sim \text{Beta}(12, 10)$ and:

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Since this is less than 0.5, we would reject H_0 and conclude $p > 1/2$.

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Case 2: $n = 20, \sum x_i = 9$. Then $p \mid x \sim \text{Beta}(10, 12)$ and:

$$P(p \leq 0.5 \mid x) = 0.668$$

Since this exceeds 0.5, we would not reject H_0 .

Bayesian Interval Estimation

Credible Intervals

Definition: Credible Interval

For a posterior distribution $\pi(\theta|x)$, an interval $[a, b]$ having posterior probability $1 - \alpha$ is called a $100(1 - \alpha)\%$ **credible interval**:

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The standard approach is an **equal-tailed** credible interval: choose a and b as the $\alpha/2$ and $1 - \alpha/2$ quantiles of the posterior distribution.

An alternative is the **highest posterior density (HPD)** credible interval: the shortest interval with the required posterior probability. For symmetric, unimodal posteriors, the equal-tailed and HPD intervals coincide.

Credible Intervals vs. Confidence Intervals

Frequentist Confidence Interval:

- The parameter θ is **fixed** (not random)
- The interval $[L(X), U(X)]$ is **random**
- Interpretation: If we repeated the experiment many times and computed an interval each time, $100(1 - \alpha)\%$ of those intervals would contain the true θ
- We do **not** say " $P(\theta \in [L, U]) = 0.95$ "

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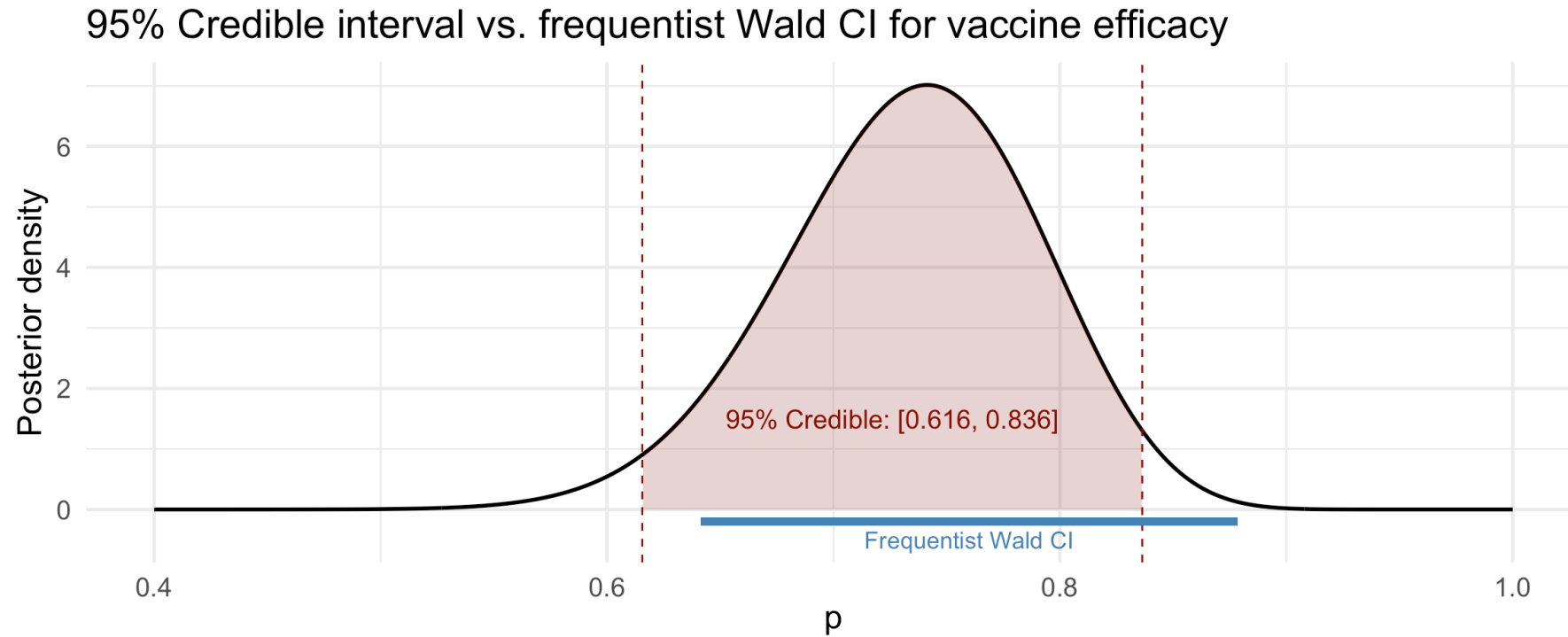
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For large samples, credible intervals and confidence intervals are often numerically similar, but their **interpretations remain fundamentally different**.

Example: Credible Intervals for Vaccine Efficacy

Continuing our vaccine example (Beta(44, 16) posterior):

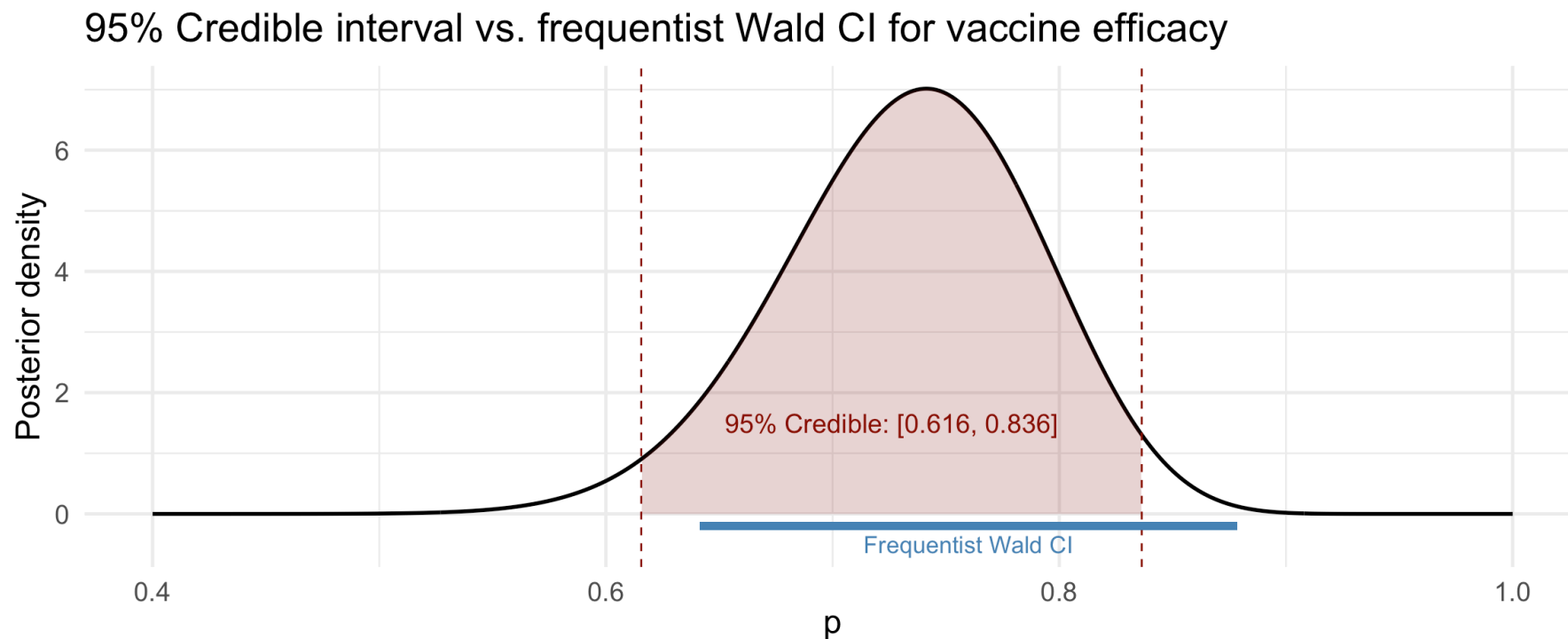
► Code



Example: Credible Intervals for Vaccine Efficacy

Continuing our vaccine example (Beta(44, 16) posterior):

► Code



Method	95% Interval	Length
Bayesian credible (Beta posterior)	[0.616, 0.836]	0.221
Frequentist Wald CI	[0.642, 0.878]	0.237

Example: Poisson Credible Interval (C&B 9.2.16)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\theta)$ with sufficient statistic $T = \sum X_i$ and prior $\text{Gamma}(\alpha, \beta)$.

The posterior is $\text{Gamma}(\alpha + t, 1/(n + 1/\beta))$.

A $100(1 - \alpha)\%$ credible interval splits α equally between the tails of this posterior.

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Using the Gamma–chi-squared relationship, if α is an integer, the credible interval can be expressed using chi-squared quantiles:

$$\left[\frac{\beta}{2(\beta n + 1)} \chi_{2(\alpha+t), 1-\alpha/2}^2, \quad \frac{\beta}{2(\beta n + 1)} \chi_{2(\alpha+t), \alpha/2}^2 \right]$$

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Compare to the frequentist confidence interval from pivoting the CDF (Lesson 8):

$$\left[\frac{1}{2n} \chi_{2t, 1-\alpha/2}^2, \quad \frac{1}{2n} \chi_{2(t+1), \alpha/2}^2 \right]$$

The Bayesian interval incorporates the prior through α and β ; the frequentist interval depends only on the data.

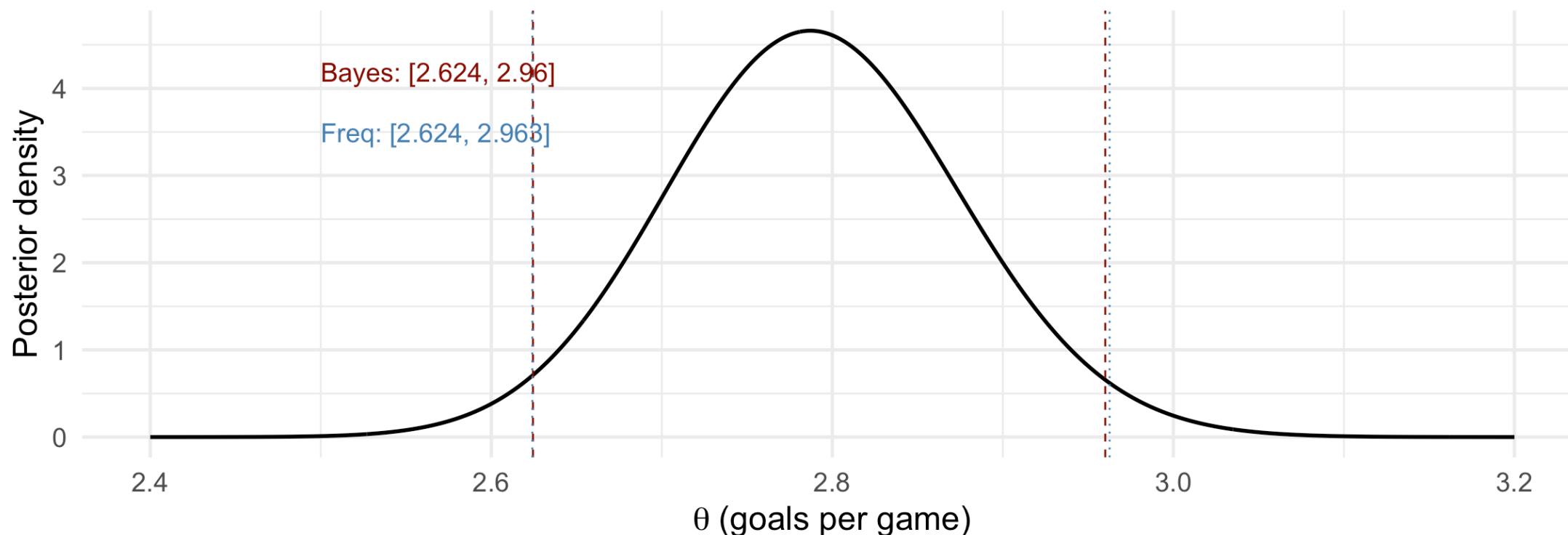
Poisson Credible Interval: Soccer Goals Example

Data from the English Premier League: $n = 380$ games, total goals $t = 1060$.

► Code

Poisson credible interval: Mean goals per game (EPL)

$n = 380$ games, total goals = 1060, Prior: $\text{Gamma}(1.5, 2)$, Posterior: $\text{Gamma}(1061.5, 0.00263)$



With $n = 380$, the Bayesian and frequentist intervals are nearly identical — the data overwhelms the prior.



Noninformative Priors and Sensitivity

Noninformative (Flat) Priors

When little prior knowledge is available, we may want a prior that has **minimal influence** on the posterior. Common choices include:

- **Beta(1, 1) = Uniform(0,1)** for a proportion — the “flat” prior
- **Beta(0.5, 0.5)** — Jeffreys’ prior for a binomial proportion
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With the **flat prior** Beta(1, 1) for binomial data, the posterior is Beta($1 + y$, $1 + n - y$), and:

$$\hat{p}_B = \frac{1 + y}{2 + n}$$

This is the **Laplace estimator** — equivalent to adding one prior success and one prior failure.

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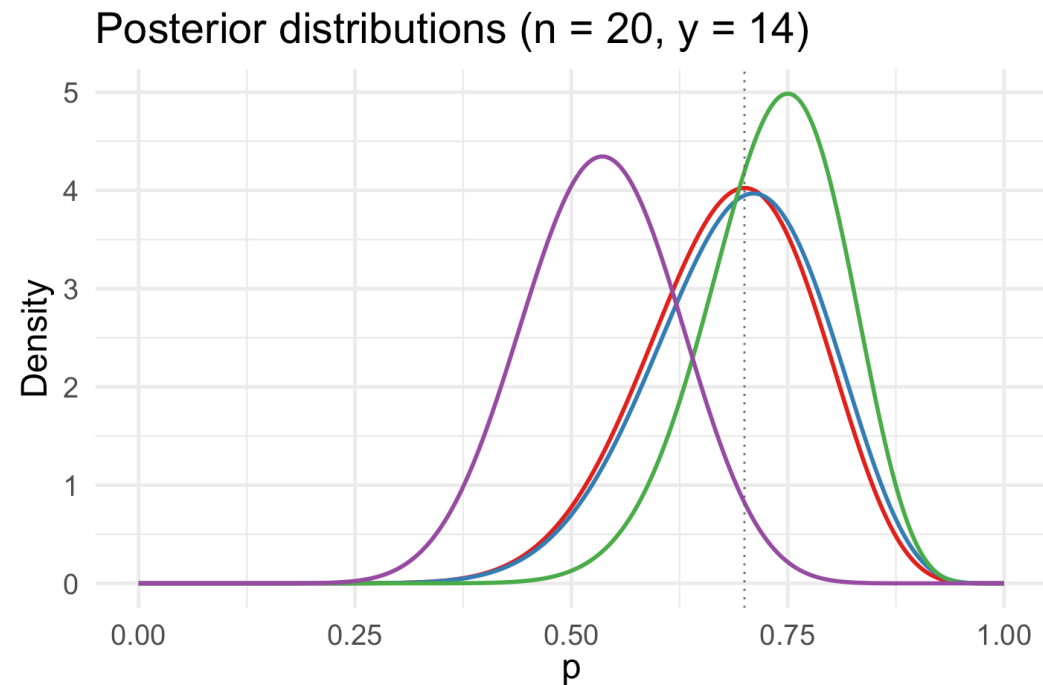
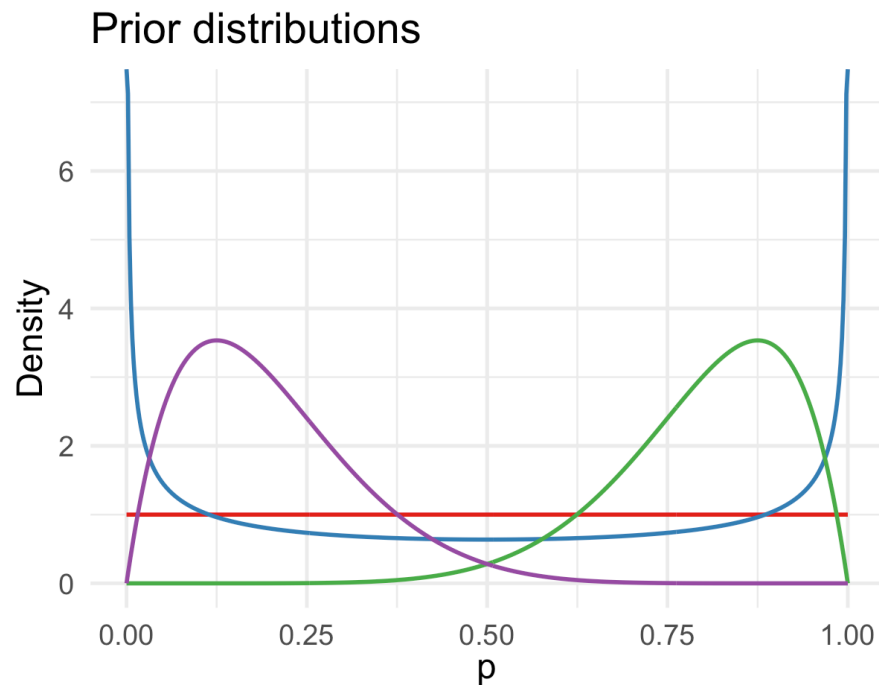
This is the **Laplace estimator** — equivalent to adding one prior success and one prior failure.

Noninformative priors let the data “speak for themselves” and produce results similar to frequentist methods. The posterior mean with a flat prior is approximately the MLE for large n .

Prior Sensitivity Analysis

A responsible Bayesian analysis should assess how sensitive the conclusions are to the choice of prior. This is called **sensitivity analysis** or **robustness analysis**.

► Code

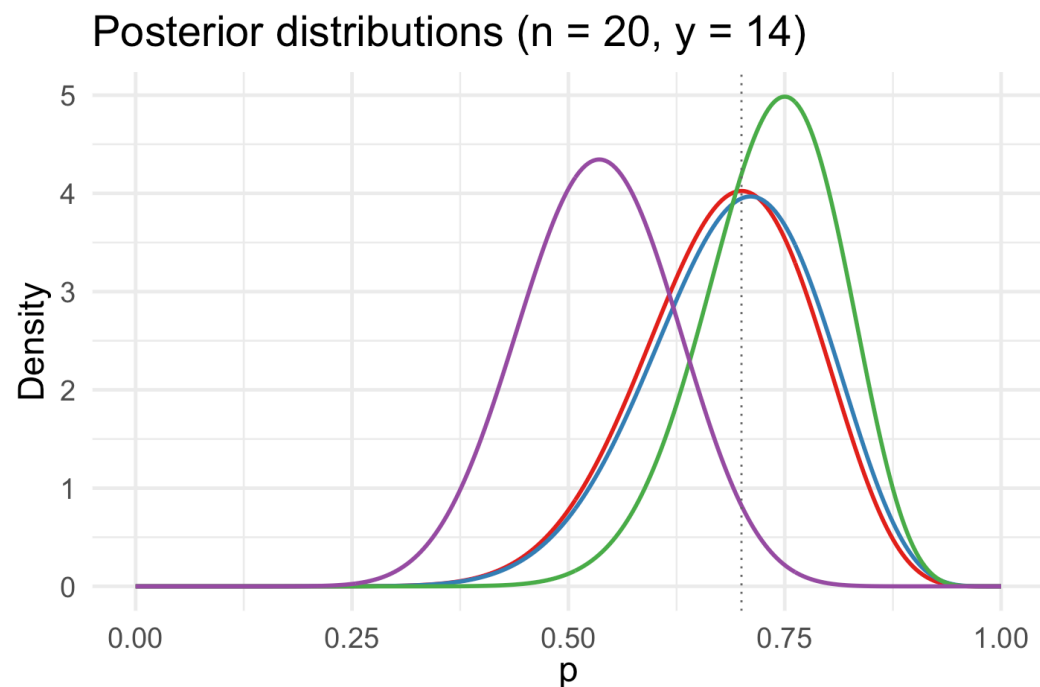
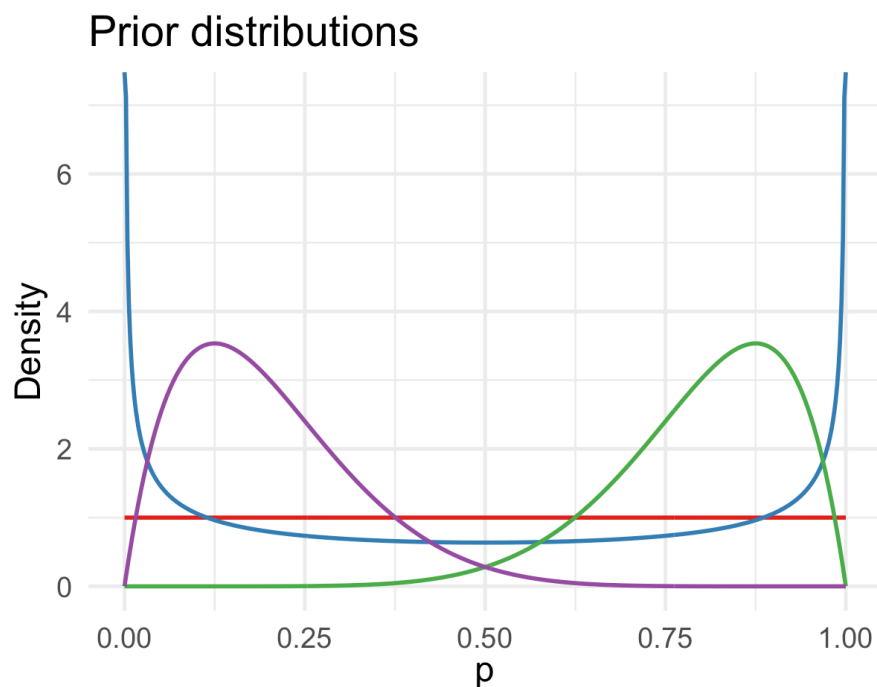


Prior — Flat: Beta(1, 1) — Jeffreys: Beta(0.5, 0.5) — Optimistic: Beta(8, 2) — Skeptic

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Even with $n = 20$, the posteriors are already converging. With larger n , the choice of prior matters even less.



Bayesian vs. Frequentist: Summary

Side-by-Side Comparison

Aspect	Frequentist	Bayesian
Parameter θ	Fixed, unknown constant	Random variable with a distribution
Probability	Long-run frequency	Degree of belief
Prior information	Not formally incorporated	Encoded in $\pi(\theta)$
Point estimate	MLE $\hat{\theta}$	Posterior mean $E(\theta x)$
Interval estimate	Confidence interval	Credible interval
Hypothesis test	P-value, rejection region	Posterior probability $P(H_0 x)$
Interpretation of 95% CI	95% of intervals from repeated samples cover θ	95% probability θ is in the interval
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As C&B state: *“There is no one best way to do statistics; some problems are best solved with classical statistics and some are best solved with Bayesian statistics.”*

Bayesian Methods in Modern Biostatistics

Clinical Trials:

- FDA guidance on Bayesian methods for medical devices (2010)
- Adaptive trial designs that update allocations based on accumulating data
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Epidemiology:

- Disease mapping and spatial modeling
- Bayesian meta-analysis combining multiple studies
- Real-time surveillance and outbreak detection

Summary and Connections

Key Takeaways

1. **Bayes' theorem** provides a principled way to update beliefs: $\text{Posterior} \propto \text{Prior} \times \text{Likelihood}$
2. **Conjugate priors** yield tractable posteriors in the same family as the prior
3. The **Bayes estimator** (posterior mean) is a weighted average of prior belief and observed data, with weights determined by relative precision
4. **Credible intervals** allow direct probability statements about parameters
5. For **large samples**, Bayesian and frequentist methods converge; differences matter most for small samples or strong priors
6. Bayesian methods offer a **coherent framework** for combining prior knowledge with data — particularly valuable in medical research where historical data and expert knowledge are abundant

Connection to Course Themes

Throughout BSTA 551, we have studied the **foundations of statistical inference**:

- **Point estimation:** MLE, method of moments, MVUE, sufficiency, Fisher information (Lessons 1–6)
- **Interval estimation:** Confidence intervals, bootstrap CIs (Lessons 7–11)
- **Hypothesis testing:** P-values, power, Neyman-Pearson, LRT (Lessons 12–17)
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Bayesian inference provides an alternative paradigm that:

- Uses the **same likelihood function** as frequentist methods
- Reaches similar conclusions for large samples (consistency of Bayes estimators)
- Offers different — sometimes more intuitive — interpretations of uncertainty
- Formalizes the incorporation of prior knowledge into analysis

References

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- Casella, G. and Berger, R. L. *Statistical Inference* (2nd ed.). Sections 8.2.2, 9.2.16.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis* (3rd ed.). CRC Press.

Questions?

Thank you!